AWS – ELB (Elastic Load Balancer) & AutoScaling (AS)

VISHWANATH M S

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AWS – ELB (Elastic Load Balancer)

TCP/IP Layer

TCP/IP Layers

Application Port No 1 to 65536
Transport TCP/UDP
Network IP address
Datalink MAC Address

Load Balancer works on the Application layer and does per PORT load Balancing

Load Balancer provides HIGH AVAILABILITY ON APPLICATION LAYER

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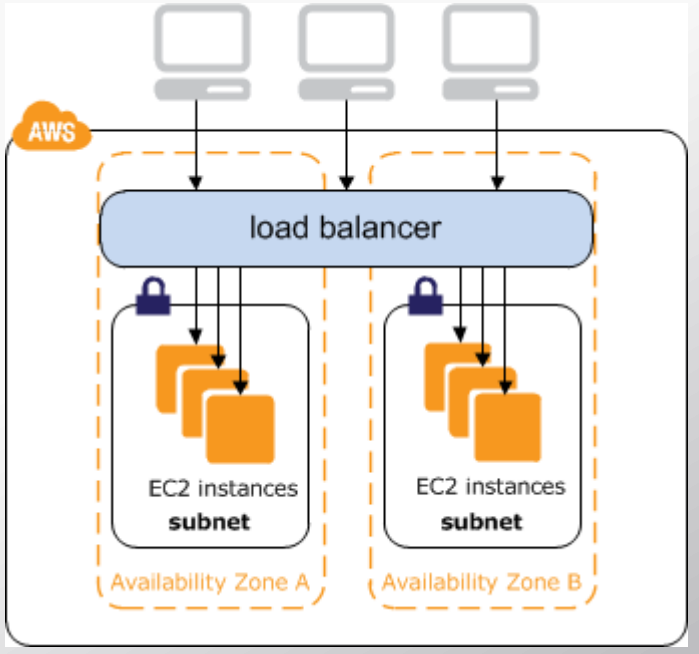
Concepts of ELB (ElasticLoadBalancer)

- LoadBalancer is used to share the Load, so that we DON'T have a single point of failure.
- AWS LoadBalancer has 2 Parts in it
  - Create Load Balancers – To configure the Front End
  - Create Target Groups -- To configure the Back End
- A LoadBalancer can also work on IPV6 public IP but only In "Application LB".
- A LoadBalancer is a request Forwarder and does not Acknowledge to the request itself to the end users.
- A LoadBalancer works WITHIN the VPC.



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Concepts of ELB (ElasticLoadBalancer)

- Multiple options are
  - Application Load Balancer
  - Network Load Balancer
  - AWS Classic Load Balancer

- Application Load Balancer
  - To perform LB on HTTP and HTTPS

- Network Load Balancer
  - To perform LB on any TCP port

- AWS Classic Load Balancer
  - Previous Generation → HTTP, HTTPS and TCP

Application Load Balancer

- **Pre-requisite**
  - We need to have minimum of 2 Subnets and these 2 subnets should be in 2 different Availability Zone.
- **Configuration**
  - Only 1 subnet can be selected in an Availability Zone
  - Also supports Ipv6 on the FrontENd of Loadbalancer

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Create Load Balancer

- 4 Sub steps
- Sub-Step1: Configure Load Balancer
- Sub-Step2: Security Group
- Sub-Step3: Configure Routing
- Sub-Step4: Register Target

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Step1: Load Balancer Configuration

- Configuration has 2 sub parts
  - Front End
    - Whether the LB has to “internet Facing” Or “Internal” facing
    - The Front End Public IP is created Automatically for “Application LB”
    - We have a option to assign an Static Public Ip (ElasticIP) for “Network LB”
  - Back End
    - For Application LB
      - We need to add a VPC that has a min of 2 Subnets in 2 Different Availability Zone.
    - For Network LB
      - We can have a min of 1 Subnet in the backend.

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Step2: Security Group

- Since LB is an individual feature, we need to specify what traffic can come in.
- Security Group option is available only in “Application LB” and NOT in “Network LB”.

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After the Creation of LB

- There is an Automatic “DNS” name created by AWS, also called as “A Record” for IPV4 or “AAAA Record” for IPv6.
- For “Application LB” the IPV4 and IPV6 address is automatically assigned.
- In case of “Network LB”, we have an option to use an Elastic IP for the LB DNS name.

Network Load Balancer

Name	DNS name	State	VPC ID	Availability Zones
LB01	LB01-c582d05294ffff04.elb.u...	active	vpc-08ed4f70	us-west-2a

Load balancer: LB01

Description | Listeners | Monitoring | Integrated services | Tags

Basic Configuration

Name	LB01
ARN	arn:aws:elasticloadbalancing:us-west-2:558457644648:loadbalancer/net/LB01/c582d05294ffff04
DNS name	LB01-c582d05294ffff04.elb.us-west-2.amazonaws.com (A Record)
State	active
Type	network
Scheme	internet-facing
IP address type	ipv4
VPC	vpc-08ed4f70
Availability Zones	subnet-e16930aa - us-west-2a IPv4 address: Elastic IP 34.214.43.191

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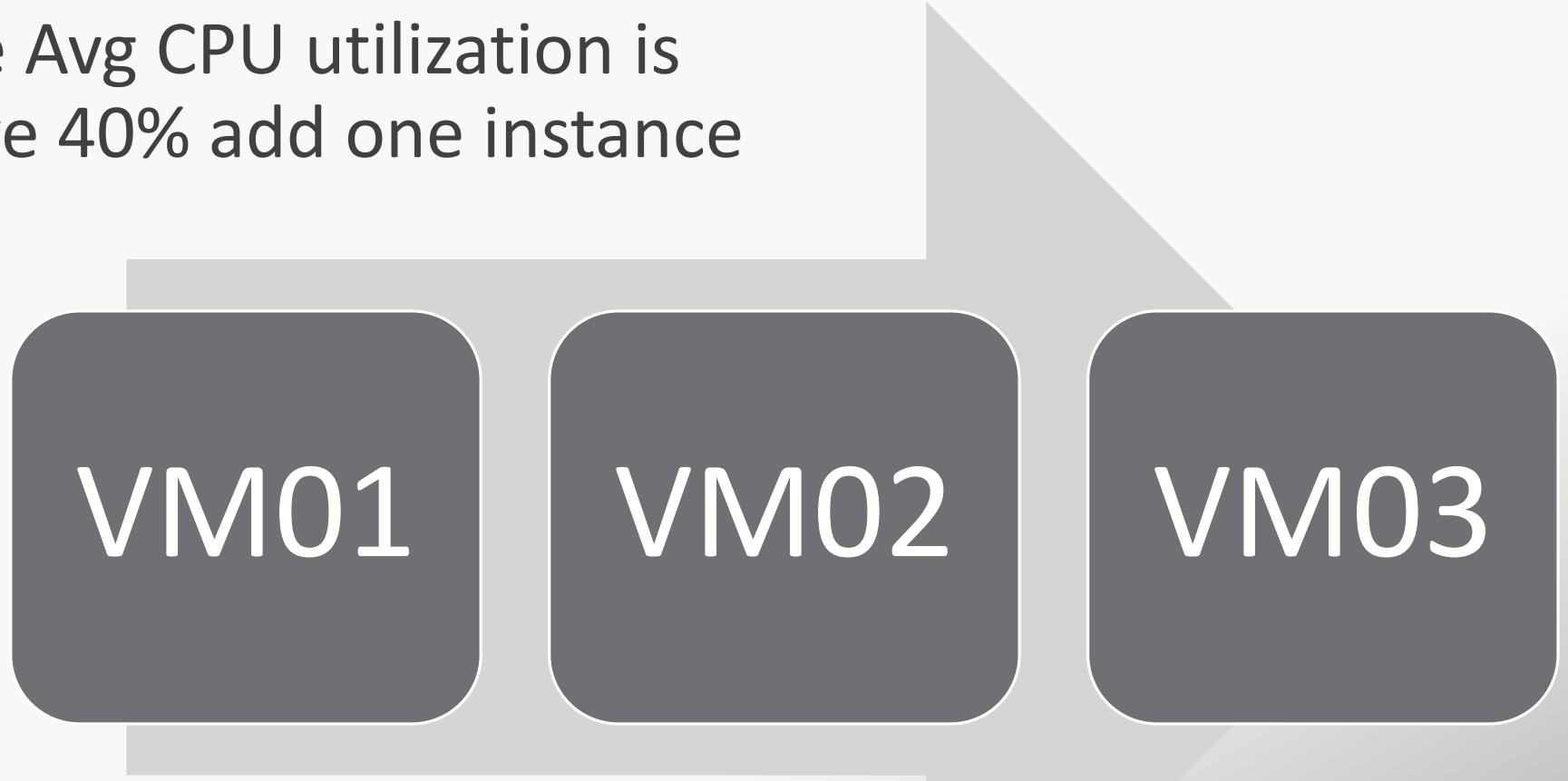
AWS – AutoScaling (AS)

AutoScaling

- It can Scale Out and Scale in
- During Scale In, the VM's are deleted in FIFO Manner (First in First Out)

Scale Out Flow

- If the Avg CPU utilization is Above 40% add one instance



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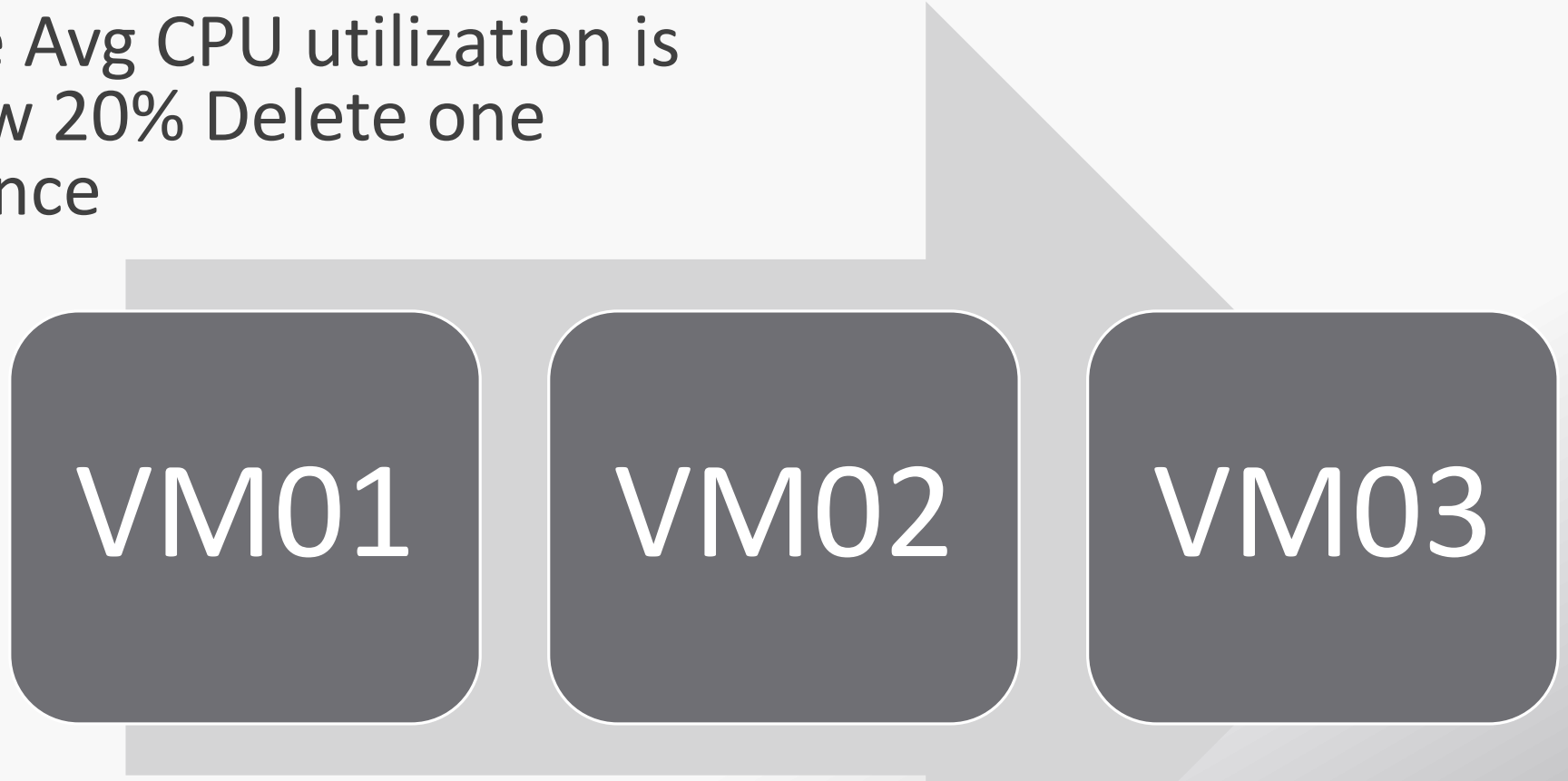
Scale IN Flow

- If the Avg CPU utilization is Below 20% Delete one instance



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Autoscaling Creation

- 2 Parts in it
  - Launch Configuration
  - Autoscaling Group

Auto Scaling Launch Configuration

- **Options Configured for an EC2**
- AMI
- Instance type
- Launch Config name with Advance Details
  - Specify whether IPV4 public is required are not on each EC2 instance
- Disks/Volume
- Security Group
- KEYS

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Autoscaling Group

- Configure Autoscaling Group Details
  - Autoscaling Group Name.
    - Initial Size of the Group
    - Network /Subnets (Select the same VPC that was selected for the SG in the Launch Configuration)
  - Under the **Advance Config**
    - **Instance Scale IN Protection**

Scaling Policy

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Deleting VM's in Autoscaling

If you delete all the VM's in the Autoscaling, Autoscale would create the min VM's as per the configuration again.

If you want to delete all the VM's in the Autoscale, then DELETE the AUTOSCALE GROUP

Autoscaling Use Case:

Scaling Website's within the same Availability Zone

Automatically scale the Website across Availability Zone.